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EVEN END SEMESTER EXAMINATION, June- 2025

Name of the Course: B.Tech
Name of the Paper : Structural Analysis- I
Time: 3 Hours

Semester : IV semester
Paper Code: TCE 402
Maximum Marks: 100

Note: i) All questions are compulsory
ii) Answer any two sub questions among a, b & c in each main question
iii) Total marks in each main question are twenty
iv) Each question carries 10 marks.

Q.1		(20 Marks)	
	a)	A simply supported beam of length 5m carries a point load of 5KN at a distance of 3m from the left end. If $E=2x10^5$ N/mm² and $I = 10^8$ mm⁴, determine the slope at the left support and deflection under the point load using Conjugate beam method.	(CO 1)
	b)	A simply supported beam of span L=8m carries a uniformly distributed load w=5kN/m over the entire span. Determine the maximum deflection and the location of maximum deflection using the moment-area method. Assume EI is constant.	(CO 1)
	c)	A fixed beam with a span of L carries a central concentrated load P. Derive the expression for the fixed-end moments at both supports and determine their values for P=50kN and L=5m	(CO 1)
Q.2		(20 Marks)	
	a)	A beam of 4m length is simply supported at the ends and carries a uniformly distributed load of 6kN/m length. Determine the strain energy stored in the beam. Take $E=2x10^5$ N/mm² and I= 1440 cm⁴.	(CO 2)
	b)	A steel bar 4cm by 4cm in section, 3m long is subjected to an axial pull of 128kN. Taking E=200 GN/m². Calculate the change in length of the bar. Also calculate the amount of energy stored in the bar during the extension.	(CO 2)
	c)	Compare the strain energies of the following two shafts subjected to the same maximum shear stress in torsion: i) A hollow shaft having outer diameter n times the inner diameter ii) A solid shaft Masses, lengths and materials of the two shafts are the same.	(CO 2)
Q.3		(20 Marks)	
	a)	A continuous beam ABCD of length 15m rests on four supports covering 3 equal spans and carries a uniformly distributed load of 1.5kN/m length. Calculate the moments and reactions at the supports. Draw the S.F and B.M diagrams	(CO 3)
	b)	Analyse the continuous beam shown in figure 1, by using three moment equation and draw the bending moment diagram.	(CO 3)

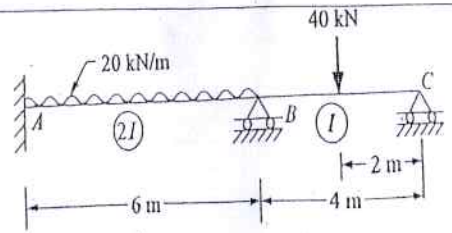

Figure 1

	c)	A continuous beam ABCD, simply supported at A,B,C and D is loaded as shown in Figure 2. Find the moments over the beam and draw B.M diagrams	(CO 3)

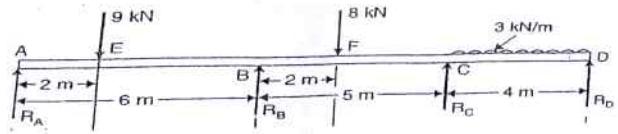

Figure 2

Q.4		(20 Marks)	
	a)	A train of concentrated loads shown in figure 3 moves from left to right on a simply supported girder of span 16m. Determine the absolute maximum shear force and bending moment developed in the beam.	(CO 4)

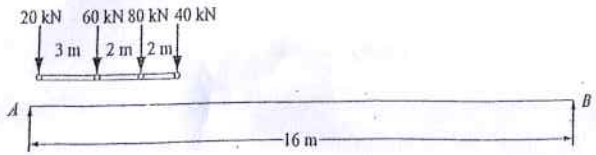

Figure 3

	b)	A train of 5 wheel loads as shown in figure 4 crosses a simply supported beam of span 24 m from left to right. Calculate the maximum positive and negative shear force values at the centre of the span and the absolute maximum bending moment anywhere in the span.	(CO 4)

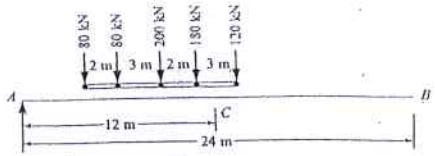

Figure 4

	c)	The system of concentrated loads shown in figure 5 rolls from left to right on the girder of span 15m, 40kN load leading. For a section 4 m from left support, determine i) Maximum bending moment ii) Maximum shear force	(CO 4)

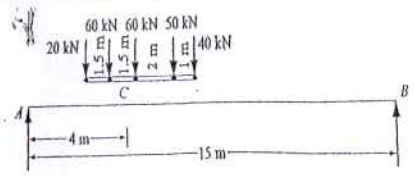

Figure 5

Q.5		(20 Marks)	
	a)	A three hinged semi-circular arch of radius R carries a uniformly distributed load of intensity w/unit length over its entire horizontal span. Determine the reactions of supports and maximum bending moment in arch.	(CO 5)
	b)	A three hinged parabolic arch having supports at different levels shown in figure 6 carries a uniformly distributed load of intensity 30kN/m over the portion left of the crown. Determine the horizontal thrust developed. Find also the bending moment, normal thrust and radial shear force developed at a section 15m from the left support.	(CO 5)
	c)	A three hinged symmetric circular arch has a span of 36m and a rise of 6m. Determine the bending moment, normal thrust and radial shear at 9m from the left support, if the arch is subjected to a uniformly distributed load of intensity 30kN/m over left half-portion and a concentrated load of 60kN at 27 m from the left springing.	(CO 5)

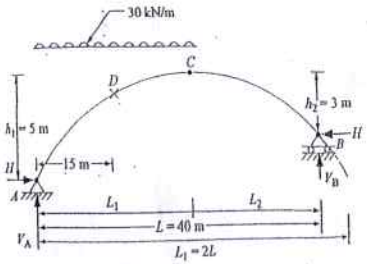

Figure 6